

Rushil Ketanbhai Vegada

(732) 467-0557 | Website: rushilvegada.com | rushilvegada@gmail.com | [Linkedin/rushilv20](https://www.linkedin.com/in/rushilv20) | [Github/rushilv20](https://github.com/rushilv20) | Raleigh, NC

EDUCATION

Master's, Computer Science

GPA: 3.7 / 4.0

North Carolina State University, Raleigh, NC

Bachelor of Technology, Computer Engineering

GPA: 3.8 / 4.0

Pandit Deendayal Energy University, Gandhinagar, India

TECHNICAL SKILLS

Programming & Frameworks: Python, JavaScript, TypeScript, ReactJS, Node.js, Express.js, Flask, Java, C++, R, Ruby, HTML, CSS

Software Development & Databases: Git, CI/CD pipelines (GitHub Actions, Jenkins), RESTful APIs, GraphQL, Microservices, Automated Testing, A/B Testing, TDD, JIRA, Agile, Scrum, MongoDB, MySQL, PostgreSQL, DynamoDB, Redis, Selenium, Postman

Cloud and DevOps: AWS, Google Cloud, Firebase, Docker, Kubernetes, Terraform, Ansible, Apache Spark and Kafka

TECHNICAL WORK EXPERIENCE

Software Engineer | Phytotron Research Facility, Raleigh, NC

Jul 2024 - Present

- Building a full-stack MERN platform with frontend (ReactJS, TypeScript, JavaScript), backend (Node.js/Express), and RESTful APIs; hosted on LAMP VM using Docker Compose and GitHub Actions CI/CD pipeline.
- Created a multi-tenant MongoDB schema to support RBAC, real-time space tracking and assignment across 100+ chambers & greenhouses, enabling automated usage-based billing and generating \$500K+ in annual transactions with integrated PDF invoicing.
- Ensured 99.9% uptime by using AWS monitoring/logging tools with PM2 & Apache reverse proxy for reliable traffic & load handling.

Software Engineer Intern | BellAir Aviation, Raleigh, NC

May 2024 - Aug 2024

- Developed a real-time business insights dashboard for BellAir, enhancing data accessibility for 80+ employees of BellAir.
- Automated data extraction & upload to AWS DynamoDB using Python scripts & automated pipelines, reducing manual efforts by 90%.
- Designed a data pipeline with AWS Lambda & Athena, exposing queryable endpoints via GraphQL APIs for real-time insights.
- Hosted the frontend on AWS Amplify and backend on Elastic Beanstalk, securing the site with custom domain and SSL from Route53.

Cloud Developer Intern | Cloud Ladder Consulting, Irving, TX (Remote)

May 2022 - Jul 2022

- Built a scalable, full-stack web-app with Flask, JavaScript, and AWS/GCP services, reducing load time by 50%.
- Enhanced location services with Google Maps API and AWS CloudFront, cutting API response time by 35%.
- Achieved 99.9% system uptime using AWS CloudTrail, Route 53, and CloudFormation for infrastructure management.

Software Developer Intern | Initialize Inc., San Ramon, CA (Remote)

Jul 2021 - Aug 2021

- Engineered an Authentication tool and Payment API using MERN stack, optimizing onboarding by 30% and improving transaction speed by 25% respectively; created a Python tool for phone verification with an accuracy by 95%.
- Conducted rigorous UI evaluations using JIRA & Excel, resolving 100+ critical bugs and ensuring design compliance across platforms.
- Streamlined and optimized RESTful APIs and microservices, achieving 20% faster data retrieval and validating enhancements with automated testing tools (Postman).

PROJECTS

KubeLine - CI/CD pipeline to deploy on Kubernetes | DevOps

- Built & containerized backend microservices using Docker & deployed on GKE with Terraform, to improve scalability & reliability.
- Integrated a CI/CD pipeline using GCP Cloud Build, Cloud Source Repository, and Artifact Registry to automate deployment.
- Utilized persistent volumes in GKE for stateful data management, ensuring reliable data storage and retrieval across deployments.

AiCurate - AI-Powered Image Curation Platform | Cloud Software Development

- Developed a scalable, full-stack platform with Python, ReactJS, and Node.js, integrating AWS services (EC2, Lambda, S3, DynamoDB, API Gateway) to support automation and secure APIs for image moderation.
- Implemented an alert system using AWS SQS, SNS, and AI Rekognition to notify admins of flagged content.
- Launched a serverless architecture utilizing AWS Lambda for content moderation and batch processing, and leveraged CloudFormation for Infrastructure as Code (IaC) deployment.

OTHER WORK EXPERIENCE

- Research Assistant** | NCSU | Built ResNet50 & CNN models (95% accuracy) for cotton fiber detection and annotated 1700+ images.
- Teaching Assistant** | NCSU | Taught "Intro to Business Analytics" to 80 students, covering Excel, Tableau, PostgreSQL, & SAS JMP.
- Research Scholar** | PDEU | Researched Fake news detection in textual data using GANs and BERT under Dr. Santosh Kumar Bharti.
- Indian Delegate** | Harvard University Asia Conference | Presented on NFT & Blockchain based solutions to tackle misinformation.
- Founder & Event Management Head** | The Network Club | Hosted the Largest Career Fair in the state history to boost networking.
- Student Scholar** | Physical Research Laboratory | Represented my state & presented on technology & space to ISRO & LIGO scientists.